

ĐẠI HỌC QUỐC GIA HÀ NỘI
TRƯỜNG ĐẠI HỌC CÔNG NGHỆ



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NGHIÊN CỨU GIẢI PHÁP TỐI ƯU
CẤU TRÚC CỦA ĐỘI HÌNH ROBOT
PHỐI HỢP ĐẨY VẬT

ĐỒ ÁN TỐT NGHIỆP ĐẠI HỌC HỆ CHÍNH QUY

Ngành: Kỹ thuật Robot

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SOA and Related Works

1.1 Inertial Parameters Estimation

To the best of our knowledge, there has been no comprehensive review dedicated specifically to center-of-mass (CoM) estimation for manipulated objects. Existing surveys primarily focus on the identification of the complete set of inertial parameters rather than CoM estimation itself. A survey of Schedlinsk [1] presents a literature review of existing methodologies on how to estimate these parameters in mechanical engineering, whereas Mavrakis and Stolkin [2] is specific to robotics. According to Mavrakis and Stolkin’s taxonomy, the methods for estimating inertial parameters can be categorized into three main groups by degree of interaction — purely visual, exploratory, and fixed-object.

The first visual methods relied on the assumption of a known and uniform object density in order to simplify the estimation problem by reducing it to the computation of the object’s volume. In [3], the author decomposed a rigid body into polyhedral components with uniform density and computed the mass, center of mass, and inertia tensor of each component before combining them to obtain the inertial properties of the complete object. The computation was achieved by converting the required three-dimensional volume integrals into two-dimensional surface integrals and one-dimensional line integrals using the divergence theorem, projection, and Green’s theorem. However, while visual perception can accurately capture an object’s shape and volume, the underlying density distribution is generally unknown, may vary throughout the object, and cannot be directly observed solely from visual information.

Exploratory and fixed-object methods, on the other hand, involve physical interaction with the object to perform the estimation task.

Tài liệu tham khảo

- [1] Carsten Schedlinski và Michael Link (2001), “A survey of current inertia parameter identification methods”. *Mechanical Systems and Signal Processing*, tập 15, số 1, tr. 189–211, doi:10.1006/mssp.2000.1345.
- [2] Han Liu et al. (2022), “Estimation and exploitation of objects’ inertial parameters in robotic grasping and manipulation: A survey”. *IEEE Transactions on Automation Science and Engineering*.
- [3] Brian Mirtich (1996), “Fast and accurate computation of polyhedral mass properties”. *Journal of Graphics Tools*, tập 1, số 2, tr. 31–50, doi:10.1080/10867651.1996.10487458.